



WORK SHEET - 17

APPLICATION OF DERIVATIVE

Q.1 (D)

$$x = 2t - t^2 \quad \dots\dots(i)$$

$$\text{and } y = t + t^2 \quad \dots\dots(ii)$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{1+2t}{2-2t}$$

Slope of tangent using the points $(1, 1)$ and $(2t - t^2, t + t^2)$

$$\frac{t + t^2 - 1}{2t - t^2 - 1} = \frac{1 + 2t}{2 - 2t}$$

$$\Rightarrow 3t^2 - 4t + 1 = 0$$

$$\Rightarrow t = 1, \frac{1}{3}$$

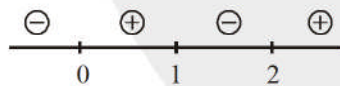
For $t = 1$, point of contact is $P(1, 2)$ and for $t = \frac{1}{3}$, point is $Q\left(\frac{5}{9}, \frac{4}{9}\right)$.

Distance between P and Q is our answer.]

Q.2 (C)

$$f(x) = \int_1^x t(t^2 - 3t + 2) dt, \quad 1 \leq x \leq 3$$

$$f'(x) = x(x-1)(x-2)$$



$\therefore$ for interval $[1, 3]$, $f \downarrow$ in $[1, 2]$ and $\uparrow$ in $[2, 3]$

$$f(x) = \int_1^x t(t^2 - 3t + 2) dt = \int_1^x (t^3 - 3t^2 + 2t) dt$$

$$= \left(\frac{t^4}{4} - t^3 + t^2 \right)_1^x = \frac{x^4}{4} - x^3 + x^2 - \frac{1}{4}$$

$$\therefore f(1) = 0, f(2) = 4 - 8 + 4 - \frac{1}{4} = -\frac{1}{4}$$



and $f(3) = \frac{81}{4} - 27 + 9 - \frac{1}{4} = 2$

$\therefore$ Range is $\left[-\frac{1}{4}, 2\right]$ **Ans.**

Q.3 (C)

Here $e = \frac{1}{2}$

$\frac{de}{dt} = -0.1$ (given)

Eccentricity of auxiliary circle = 0

$\Rightarrow \int_{1/2}^0 de = -0.1 \int_0^T dt$ T is time at which it will touch the auxiliary circle.

$\Rightarrow -\frac{1}{2} = -0.1 (T - 0)$ $\therefore T = 5$ sec.

Q.4 (B)

$f(g(x^2 - 2kx + 3)) > f(g(k - 3)) \forall x \in \mathbb{R}$

$\Rightarrow g(x^2 - 2kx + 3) < g(k - 3) \forall x \in \mathbb{R}$

$\Rightarrow x^2 - 2kx + 3 > k - 3 \forall x \in \mathbb{R}$

$\Rightarrow x^2 - 2kx + 6 - k > 0 \forall x \in \mathbb{R}$

$D < 0 \Rightarrow 4k^2 - 4(6 - k) < 0$

$\Rightarrow k^2 + k - 6 < 0$

$\Rightarrow (k + 3)(k - 2) < 0$

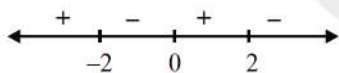
$\Rightarrow k \in (-3, 2)$

$\therefore$ Number of integral values of k are 4 i.e. $-2, -1, 0, 1$. **Ans.**

Q.5 (C)

We have

$f'(x) = e^{\frac{-x^4}{4}} \cdot (4 - x^2) \cdot 2x = 2e^{\frac{-x^4}{4}} \cdot (2 - x)(2 + x) \cdot x$



sign of $f'(x)$.

Now verify alternative.

Q.6 (D)

Given $f'(1) = 1$ and $f(2x) = f(x)$

$\Rightarrow 2 \cdot f'(2x) = f'(x)$

Putting $x = 1$, $f'(2) = \frac{f'(1)}{2} = \frac{1}{2}$

Putting $x = 2$, $f'(4) = \frac{f'(2)}{2} = \frac{1}{4}$

Now applying LMVT for $y = f'(x)$ in $[2, 4]$, we get

$$f''(c) = \frac{f'(4) - f'(2)}{2} = \frac{\frac{1}{4} - \frac{1}{2}}{2} = -\frac{1}{8}$$

Q.7 (C)

Given curve is $x^2y = c^3$.

Hence, $\frac{dy}{dx} = -\frac{2y}{x}$

Therefore, equation of tangent at (x_1, y_1) is

$$y - y_1 = -\frac{2y_1}{x_1}(x - x_1)$$

at x-axis $y = 0$, gives $x = \frac{3x_1}{2} = a$

at y-axis $x = 0$, gives $y = 3y_1 = b$

Now $a^2b = \frac{9x_1^2}{4} \cdot 3y_1 = \frac{27}{4}x_1^2y_1 = \frac{27}{4}c^3$ **Ans.**

Q.8 (C)

(A) True, every continuous function is bounded on a closed interval.

(B) True, by intermediate value theorem.

(C) Not true, extremum can occur at boundary also.

(D) True. By L.M.V.T. there exists a point $c \in (a, b)$ where $f'(c) = \frac{f(b) - f(a)}{b - a}$, a clearly positive value.]

Q.9 (A)

Given $y = \frac{x^3}{(x-a)^2} \Rightarrow y' = \frac{x^2(x-3a)}{(x-a)^3}$

$\begin{array}{c} + \quad - \quad + \\ \leftarrow \quad \quad \rightarrow \\ 0 \quad \quad 3a \end{array}$ (when $a > 0$)

sign of $\frac{dy}{dx}$

$\begin{array}{c} - \quad + \quad + \\ \leftarrow \quad \quad \rightarrow \\ 3a \quad 0 \end{array}$ (when $a < 0$)

sign of $\frac{dy}{dx}$

$\frac{dy}{dx} = 0$ gives $x = 0$ or $x = 3a$

and $\frac{dy}{dx}$ changes sign at $x = 3a$

Hence y is minimum at $x = 3a$ and its value is

$$y = \frac{(3a)^3}{(2a)^2} = \frac{27a^2}{4a^2} = \frac{27}{4}a \quad]$$

Q.10 (B)

$$f(x) = \frac{e^{-x}}{\sqrt{k+8x^2}}$$

$$\begin{aligned} f'(x) &= \frac{\sqrt{k+8x^2}(-e^{-x}) - e^{-x} \frac{16x}{2\sqrt{k+8x^2}}}{(k+8x^2)} \\ &= \frac{(k+8x^2)(-e^{-x}) - e^{-x}8x}{(k+8x^2)^{3/2}} \\ &= \frac{-e^{-x}(8x^2+8x+k)}{(k+8x^2)^{3/2}} \end{aligned}$$

for $f(x)$ always to be decreasing $f'(x) \leq 0 \forall x \in \mathbb{R}$

$$\Rightarrow 8x^2 + 8x + k \geq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow D \leq 0 \Rightarrow 16 - 32k \leq 0 \Rightarrow k \geq 2$$

$\therefore$ Least integral value of k is 2. **Ans.**

Q.11 (C)

Q.12 (B)

$$f(x) = x^3 - 6 \sin \theta \cdot x^2 + (2 \sin^2 \theta + q)x - 1$$

$$f'(x) = 3x^2 - 12 \sin \theta \cdot x + 2 \sin^2 \theta + q$$

For exactly one tangent to be drawn to the graph of $y = f(x)$, parallel to $y = x$.

$$\therefore f'(x) = 1 \text{ should have one root}$$

$$D = 0$$

$$\Rightarrow q = 1 + 10 \sin^2 \theta$$

$\therefore$ largest value of q is 11. **Ans.**

(ii) Smallest value of $q = 1$ when $\sin \theta = 0$

$$\text{Now, } f(x) = x^3 + x - 1$$

$$f(2) = 9 \Rightarrow f^{-1}(9) = 2$$

$$\left. \frac{d}{dx} (f^{-1}(x)) \right|_{x=9} = \frac{1}{f'(2)} = \frac{1}{13} \quad \{ \because f'(x) = 3x^2 + 1 \}$$

Q.13 (C)

Q.14 (B)

Q.15 (A)



Clearly, $x_1 = -x_2$ and $P'(x) = -P'(-x)$
 $\Rightarrow x_1 = 1, x_2 = -1$ $P'(0) = 0$
 $\therefore P'(x) = A(x^2 - 1)x$

$$P(x) = A\left(\frac{x^4}{4} - \frac{x^2}{2}\right) + c. \text{ also } P(0) = 1 \Rightarrow c = 1$$

Now, $y = 1 = \left(\frac{x^4}{4} - \frac{x^2}{2}\right) + 1 \Rightarrow x = \pm\sqrt{2}, 0$

$$\therefore \int_{-\sqrt{2}}^{\sqrt{2}} 1 - \left(A\left(\frac{x^4}{4} - \frac{x^2}{2}\right) + 1\right) dx = \frac{8\sqrt{2}}{15} \Rightarrow A = 2$$

$$\therefore P(x) = 2\left(\frac{x^4}{4} - \frac{x^2}{2}\right) + 1$$

$$\therefore \text{Minimum value} = P(1) = \frac{1}{2}$$

Inflexion point at $P''(x) = 0$

$$(3x^2 - 1) = 0 \Rightarrow x = \pm\frac{1}{\sqrt{3}}$$

Also, $\lim_{x \rightarrow 0} \frac{2\left(\frac{x^4}{4} - \frac{x^2}{2}\right) + 1 - (2ax^2 + 2c)}{x^2} = -2a + b$, where $g(x) = ax^2 + bx + c$

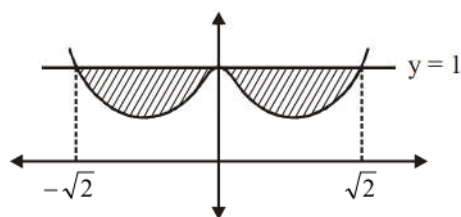
$$1 + 2c = 0 \Rightarrow c = -\frac{1}{2}$$

$$-2a - 1 = -2a + b \Rightarrow b = -1$$

Now, $g_{\max}$ at $x = -1$, $\frac{-b}{2a} = -1 \Rightarrow a = -\frac{1}{2}$

$$\therefore g(x) = \frac{-1}{2}x^2 - x - \frac{1}{2}$$

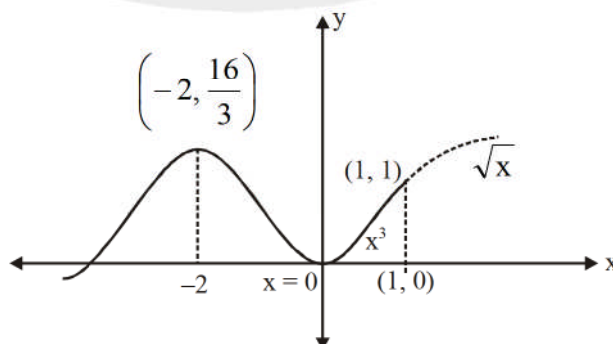
Product of roots = 1.



Q.16 (ABC)

$$f(x) = \frac{x^3}{3} - 4x \Rightarrow f'(x) = 0 \text{ gives } x^2 - 4x = 0 \Rightarrow x = \pm 2$$

$$f'(x) = \begin{cases} \frac{1}{2\sqrt{x}}, & x \geq 1 \\ 3x^2, & 0 \leq x < 1 \\ x^2 - 4, & x < 0 \end{cases}$$



$$f'(1^+) = \frac{1}{2}, f'(1^-) = 3$$

$$f'(0^+) = 0, f'(0^-) = -4$$

Hence, function is not differentiable at $x = 0$ and 1 .

Q.17 (BCD)

$$g(x) = \int_2^6 f(2x+t) dt$$

$$\text{Let } 2x+t=y \Rightarrow dt=dy$$

$$g(x) = \int_{2x+2}^{2x+6} f(y) dy$$

$$g'(x) = (f(2x+6) \cdot 2 - f(2x+2) \cdot 2)$$

$$g'(x) = 2(f(2x+6) - f(2x+2))$$

$$g'(0) = 2(f(6) - f(2)) = 2(8 - 3) = 10$$

$f(x)$ has local minimum at $x = 2$

$$\therefore f'(2) = 0$$

$$(C) \quad \lim_{x \rightarrow 0} \frac{g(f(x+2)-3) - g(0)}{x} = \lim_{x \rightarrow 0} \frac{g'(f(x+2)-3) \cdot f'(x+2)}{1}$$

$$= g'(f(2)-3) f'(2)$$

$$= g'(0) f'(2) = 0$$

$$(D) \quad \lim_{x \rightarrow 1} \frac{f(6x) - f(6)}{x-1} = \lim_{x \rightarrow 0} \frac{6f(x+6) - 16f(x+2)}{x}$$

$$\lim_{x \rightarrow 1} \frac{6f'(6x)}{1} = \lim_{x \rightarrow 0} \frac{6f'(x+6) - 16f'(x+2)}{1}$$

$$6f'(6) = 6f'(6) - 16f'(2)$$

$$6f'(6) = 6f'(6) \quad \text{Ans.}$$

Q.18 (ACD)

$$\text{Clearly } f(x) = 2 \Rightarrow f(x) - 2 = 0$$

$$\therefore f(x) - 2 = (x-1)^2 (x-4)^3$$

$$\therefore f(x) = (x-1)^2 (x-4)^3 + 2$$

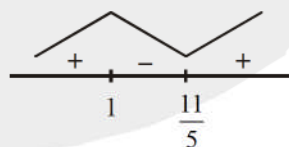
$$\begin{aligned} \therefore f'(x) &= 2(x-1)(x-4)^3 + 3(x-1)^2(x-4)^2 \\ &= (x-1)(x-4)^2 [2(x-4) + 3(x-1)] \\ &= (x-1)(x-4)^2 (5x-11) \end{aligned}$$

$$\Rightarrow x = \frac{11}{5} \text{ is a point of local minimum.}$$

$$f(2) = -6, \quad f(3) = -2$$

$$\int_2^6 f(x) dx = \int_2^6 ((x-1)^2 (x-4)^3 + 2) dx$$

$$\text{Put } x-4 = t$$



$$= \int_{-2}^2 (t^3(t+3)^2 + 2) dt = \int_{-2}^2 (t^5 + 9t^3 + 6t^4 + 2) dt = 2 \int_0^2 (6t^4 + 2) dt = 2 \left(\frac{6 \cdot 32}{5} + 4 \right) = \frac{424}{5}$$

Q.19 (BD)

Let roots of $x^3 - lx^2 + mx - 27 = 0$ are a, b , and c

$$\therefore a + b + c = l \quad \dots\dots(i)$$

$$ab + bc + ca = m \quad \dots\dots(ii)$$

$$abc = 27 \quad \dots\dots(iii)$$

for (ii)/(iii)

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{m}{27}$$

Applying A.M. $\geq$ G.M.

$$\therefore \frac{a+b+c}{3} \geq (abc)^{1/3} \Rightarrow l \geq 9$$

$$\text{and } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \left(\frac{1}{abc} \right)^{1/3} \Rightarrow \frac{m}{3 \cdot 27} \geq \frac{1}{3} \Rightarrow m \geq 27$$

$$\therefore l_{\min} = 9, \quad m_{\min} = 27$$

Q.20 [2]

$$\int_0^{t^2} f^2(x) dx + \frac{1}{2} t^6 \int_0^{\pi/2} \sin^3 x dx = \int_0^{t^2} 2x f(x) dx$$

$$\int_0^{t^2} f^2(x) dx + \frac{t^6}{2} \cdot \frac{2}{3} = \int_0^{t^2} 2x f(x) dx$$

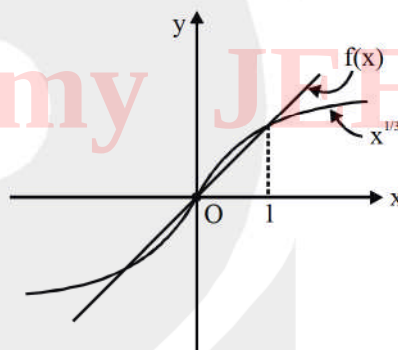
$$f^2(t^2) \cdot 2t + 2t^5 = 2t^2 f(t^2) \cdot 2t$$

$$f^2(t^2) + t^4 - 2t^2 f(t^2) = 0 \Rightarrow (f(t^2) - t^2)^2 = 0$$

$$\Rightarrow f(t^2) = t^2$$

$$\Rightarrow f(x) = x \quad \forall x \geq 0$$

From the graph, it is clear that $x = 0$ and $x = 1$ are only two solutions.



Q.21 3

For a positive point of maximum both distinct roots of $f'(x) = 0$ should be greater than zero.

$$\therefore f'(x) = x^2 - 2x + a$$

$\therefore$ both distinct greater than zero

$$(i) D > 0 \Rightarrow 4 - 4a > 0 \Rightarrow a < 1$$

$$(ii) f'(0) > 0 \Rightarrow a > 0$$

$$(iii) \frac{-(-1)}{2 \cdot \frac{1}{3}} = \frac{3}{2} > 0$$

$$\therefore a \in (0, 1)$$

$$\therefore a \in (0, 1) \text{ satisfying } 3x^2 - (b+1)x + b(b-2) < 0$$

$$\therefore b(b-2) \leq 0 \Rightarrow 0 \leq b \leq 2$$



$$\begin{aligned}\text{and} \quad & 3 - (b + 1) + b(b - 2) \leq 0 \\ \Rightarrow \quad & b^2 - 3b + 2 \leq 0 \Rightarrow 1 \leq b \leq 2 \\ \therefore \quad & b \in [1, 2] \\ \therefore \quad & p + q = 3 \quad \text{Ans.}\end{aligned}$$

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